

Dynamics of a 3-RRR Spherical Parallel Mechanism Based on Principle of Virtual Powers

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Abstract— Some recursive matrix relations for the geometric analysis, kinematics and dynamics of a 3-RRR symmetric spherical Agile Wrist parallel mechanism are established. The manipulator prototype is a three-degree-of-freedom mechanical system with three parallel legs. Controlled by concurrent torques, which are generated by some electric motors, three active elements of the robot have three independent rotations. Supposing that the position and the rotation motion of the moving platform are known, an inverse dynamic problem is developed using the principle of virtual powers to compute actuating couples. Finally, some matrix relations and some graphs for the torques and the powers of the actuators are determined¹.

Keywords: dynamics, parallel robot, virtual powers

I. Introduction

Parallel manipulators are closed-loop mechanical structures presenting very good performances in terms of accuracy, rigidity and ability to manipulate large loads. These mechanisms consist of two main bodies coupled via numerous legs. One body is arbitrarily designated as fixed, while the other is regarded as movable, and hence they are respectively called *base* and *moving platform*.

Typically, a parallel mechanism is said to be *symmetrical* if it satisfies the following conditions: the number of legs is equal to the number of degrees of freedom of the moving platform, one actuator controls every limb and the location and the number of actuated joints in all the limbs are the same (Tsai [1]).

The parallel robots can be equipped with hydraulic or pneumatic actuators. They have a robust construction and can move bodies of considerable masses and dimensions with high speeds. This is why the mechanisms, which produce a translation or spherical motion to a platform, are based on the concept of parallel manipulator.

For two decades, parallel manipulators attracted to the attention of more and more researches that consider them as valuable alternative design for robotic mechanisms [2], [3]. Compared with the serial robots, parallel mechanisms have some special characteristics: greater structural rigidity, better orientation accuracy, stable functioning, larger dynamic charge capacity and suitable position of the actuating systems. On the other hand, parallel kinematics machines offer essential advantages over serial

robots: lower moving masses, higher natural frequencies, simpler modular mechanical construction and possibility to mount all actuators at or near the fixed base. However, most existing parallel robots have limited and complicated workspace volume with singularities and highly non-isotropic input-output relations [4].

Recently many efforts have been assigned to kinematics and dynamic analysis of fully parallel manipulators. Many companies have developed them as high precision machine tools. Among these, the class of manipulators known as Stewart-Gough platform focused great attention (Stewart [5]; V. Parenti-Castelli and R. Di Gregorio [6]; Merlet [7]). They are used in flight simulators, pointing devices, and more recently, for Parallel Kinematic Machines. The Star parallel manipulator (Hervé and Sparacino [8]) and the Delta parallel robot (Clavel [9]; Staicu and Carp-Ciocardia [10]; Tsai and Stamper [11]) are equipped with three motors and train on the mobile platform in a three degree of freedom general translation motion. Angeles [4], Gosselin and Gagné [12] analysed the direct kinematics, dynamics and singularity loci of the Agile Wrist prototype that presents three concurrent rotations

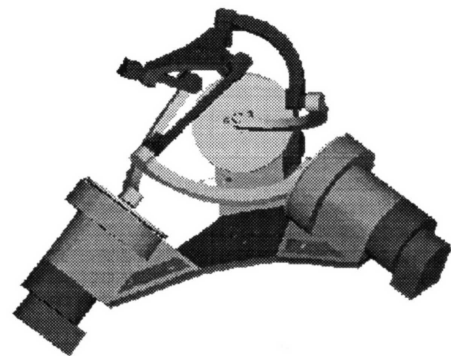


Fig. 1 Spherical parallel robot

While the kinematics has been studied extensively during the last two decades, fewer papers can be focused on the dynamics of parallel architectures.

The dynamics analysis of parallel robots is usually implemented through analytical methods in classical mechanics [13] in which projection and resolution of equations on the reference axes are written in a large number of cumbersome, scalar relations and the solutions

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are rendered by large scale computation together with time consuming computer codes.

Meanwhile, quite few of special approaches have been conducted for dynamic modelling of specific parallel mechanism configurations. Sorli et al. [14] conducted the dynamics modelling for Turin parallel manipulator, though the mechanism has three identical legs, it has 6-DOFs. Geng et al. [15] developed Lagrange's equations of motion under some simplifying assumptions regarding the geometry and inertia distribution of the manipulator. Dasgupta and Mruthyunjaya [16] used the Newton-Euler approach to develop closed-form dynamic equations of Stewart platform, considering all dynamic and gravity effects as well as viscous friction at joints.

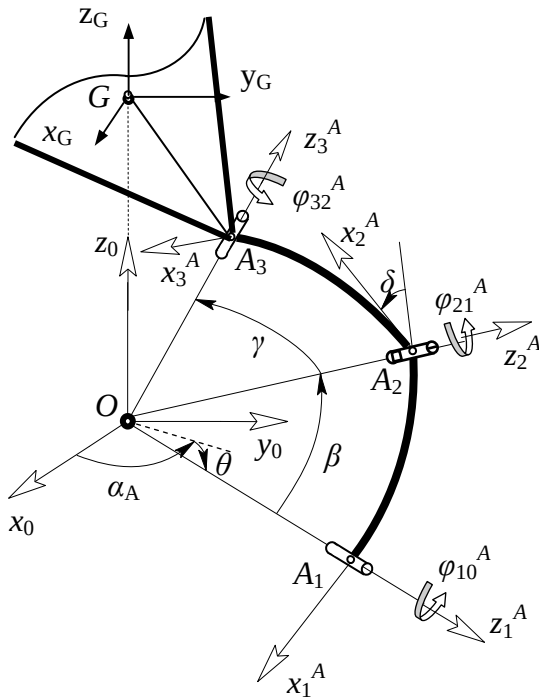


Fig. 2 Kinematical scheme of leg A of the mechanism

In the present paper we focus our attention on a recursive matrix method, which is adopted to derive the kinematics and dynamics equations of a 3-RRR symmetric spherical parallel three-degrees-of-freedom robot (fig.1).

II. Inverse geometric model

Spherical parallel manipulators provide high stiffness and accuracy due to their mechanical arrangement in which all the actuators are fixed to the base and the end-effector is supported by three kinematical chains.

Parallel spherical wrists are used in technical applications where it is desired to orient a rigid body at high speed. Accuracy and precision in the execution of the task of the wrist are essential since the robot is intended to operate on fragile objects; where positioning errors of the tool could end in costly damage. Examples of such

applications include the orientation of a camera at high velocity for the tracking of fast objects.

This robot has a structure with the axes of all nine revolute joints concurring in a common *centre O of rotation* of the device. Let $Ox_0y_0z_0(T_0)$ be a fixed Cartesian frame, about which the manipulator moves. It has three legs of known size and mass (fig.2). The wrist architecture consists of two main elements, the base and the moving platform, which is free to undergo arbitrary rotations with respect to the centre of three identical serial legs, each of these composed of two links coupled each other by means of a revolute joint.

To simplify the graphical image of the kinematical scheme of the mechanism, in the follows we will represent the intermediate reference systems by only two axes, so as one proceeds in most of robotics papers [1], [3], [4], [7]. The z_k axis is represented, of course, for each component element T_k . It is noted that the relative rotation with $\varphi_{k,k-1}$ angle of T_k body must be always pointing about the direction of z_k axis. The first element T_1 of leg A, one of the three driving parts of the robot, is called *proximal link*. It is a homogenous rod, rotating about the axis Oz_1^A with the angular velocity $\omega_{10}^A = \dot{\varphi}_{10}^A$ and the angular acceleration $\varepsilon_{10}^A = \ddot{\varphi}_{10}^A$. It has the radius r , masse m_1 and tensor of inertia $\hat{J}_1$. The *distal link* T_2 of radius r is connected to the $A_2x_2^Ay_2^Az_2^A$ frame and has a relative rotation with the angle φ_{21}^A and angular velocity $\omega_{21}^A = \dot{\varphi}_{21}^A$; it has the masse m_2 and the inertia tensor $\hat{J}_2$.

The platform of the robot is a three arms star of length $l = r\sqrt{(2 + \sin^2 \delta)/3}$, mass m_3 and tensor of inertia $\hat{J}_3$, which rotates with an angular velocity $\omega_{32}^A = \dot{\varphi}_{32}^A$ with respect to the neighbouring body. The link angles of the manipulator are assumed to be the same on each of the legs connecting the end-effector to the base and are noted β , for the proximal link and γ for the distal link. Moreover, angles $\alpha_A = 0$, $\alpha_B = 2\pi/3$, $\alpha_C = -2\pi/3$, $\delta = \pi/6$ and $\sin \theta = 1/\sqrt{3}$, $\cos \theta = \sqrt{2/3}$ give the geometry of the base and the initial position of the manipulator arm. Each leg of the Agile Wrist is a serial 3-RRR symmetric spherical wrist: the first joint couples the base with the proximal link, the second joint couples the proximal link with the distal link and the last one couples the distal link with the end-effector. The serial wrist is orthogonal, i.e., its neighbouring revolute axes lay out

at $\beta = \pi/2$, $\gamma = \pi/2$.

Let us consider the rotation angles $\varphi_{10}^A, \varphi_{10}^B, \varphi_{10}^C$ of the three actuators A_1, A_2, A_3 , the variables that give the instantaneous position of the mechanism. In the inverse geometric problem however, it can be considered that the three Euler angles $\alpha_1, \alpha_2, \alpha_3$ give the absolute orientation of the mobile platform. Pursuing the leg A along the track $OA_1A_2A_3$, one obtains the successive transformation matrices (Coiffet[3]; Staicu [17]):

$$a_{10} = a_{10}^{\theta} \theta_1 a_{\alpha}^A, \quad a_{21} = a_{21}^{\theta} \theta_2 a_{\delta}^A, \quad a_{32} = a_{32}^{\theta} \theta_1, \quad (1)$$

with

$$\theta_1 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad \theta_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad a_{\theta} = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$a_{\alpha}^A = \begin{bmatrix} \cos \alpha_A & \sin \alpha_A & 0 \\ -\sin \alpha_A & \cos \alpha_A & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad a_{\delta}^A = \begin{bmatrix} \cos \delta & \sin \delta & 0 \\ -\sin \delta & \cos \delta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$a_{k,k-1}^{\varphi} = \begin{bmatrix} \cos \varphi_{k,k-1}^A & \sin \varphi_{k,k-1}^A & 0 \\ -\sin \varphi_{k,k-1}^A & \cos \varphi_{k,k-1}^A & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad a_{k0} = \prod_{j=1}^k a_{k-j+1,k-j} \quad (k=1,2,3). \quad (2)$$

Some analogous relations can be written for the other two legs B and C of the mechanism.

For the inverse geometric analysis, the position of an arbitrary end-point $P(x_0^P, y_0^P, z_0^P)$ is treated as known and the goal is to find the joint variables $\varphi_{10}^A, \varphi_{10}^B, \varphi_{10}^C$ that yield the given location of the tool. If the aim is to generate a sequence of points to move the tool along an arc, care must be taken to avoid branch switching during motion, which may cause inefficient or impossible robot motions. The three rotation angle $\varphi_{10}^A, \varphi_{10}^B, \varphi_{10}^C$ of the actuators A_1, B_1, C_1 are the joint variables that give the input vector $\bar{\varphi}_{10} = [\varphi_{10}^A \quad \varphi_{10}^B \quad \varphi_{10}^C]^T$ of the instantaneous position of the mechanism. But, the objective of the inverse geometric problem is to find the vector $\bar{\varphi}_{10}$ and the position of the mechanism with the given orientation of the moving platform.

Let us suppose, for example, that the absolute motion of the platform is a rotation around the fixed point O, expressed by the Euler angles

$$\alpha_l = \alpha_l^* (1 - \cos \frac{2\pi}{3} t), \quad (l=1,2,3). \quad (3)$$

Representing the orientation of the platform in the fixed frame, successive relative rotations around the movable

axes Ox, Oy, Oz lead to the commonly known rotation matrix

$$a = a_3 a_2 a_1. \quad (4)$$

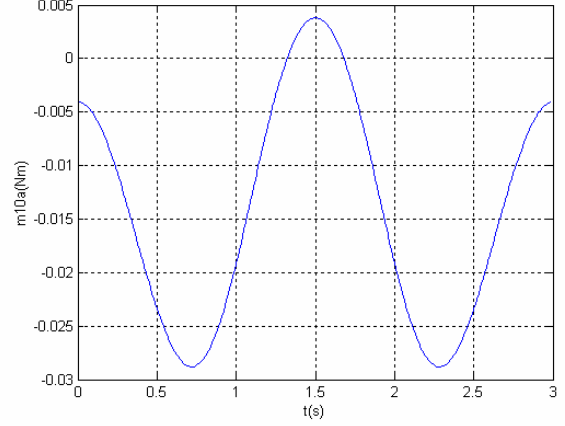


Fig. 3 Torque m_{10}^A of first actuator

Geometric conditions of rotation for the platform are given by the following identities

$$a_{30}^{\circ T} a_{30} = b_{30}^{\circ T} b_{30} = c_{30}^{\circ T} c_{30} = a, \quad (5)$$

with, for example,

$$a_{30}^{\circ} = a_{\alpha}^{AT} a_{\theta}^T \theta_1^T \theta_2^T a_{\delta}^T \theta_1^T. \quad (6)$$

From these relations, one obtains the real time evolution of nine characteristic angles $\varphi_{10}^A, \varphi_{21}^A, \varphi_{32}^A, \varphi_{10}^B, \varphi_{21}^B, \varphi_{32}^B, \varphi_{10}^C, \varphi_{21}^C, \varphi_{32}^C$.

III. Velocities and accelerations

The kinematics of compounding elements of each leg (for example the first leg A) are characterized by three skew-symmetric matrices given by the recurrence relations (Staicu [18]):

$$\tilde{\omega}_{k0}^A = a_{k,k-1} \tilde{\omega}_{k-1,0}^A a_{k,k-1}^T + \omega_{k,k-1}^A \tilde{u}_3, \quad (k=1,2,3), \quad (7)$$

which are associated to the absolute angular velocities

$$\tilde{\omega}_{k0}^A = a_{k,k-1} \tilde{\omega}_{k-1,0}^A + \omega_{k,k-1}^A \tilde{u}_3, \quad \omega_{k,k-1}^A = \dot{\varphi}_{k,k-1}^A. \quad (8)$$

The following relation gives the velocity the joint A_k :

$$\tilde{v}_{k0}^A = a_{k,k-1} \tilde{v}_{k-1,0}^A + a_{k,k-1} \tilde{\omega}_{k-1,0}^A \tilde{r}_{k,k-1}^A, \quad \tilde{v}_{k,k-1}^A = \tilde{0}. \quad (9)$$

Knowing the rotation motion of the platform by the relations (3), we develop the inverse kinematical problem and determine the velocities $\tilde{v}_{k0}^A, \tilde{\omega}_{k0}^A$ and accelerations

$\tilde{\gamma}_{k0}^A, \tilde{\varepsilon}_{k0}^A$ of each of the moving links.

Equations of geometrical constraints (5) can be derivate with respect to time to obtain the following *matrix conditions of connectivity* established for the relative angular velocities:

$$\omega_{10}^A \bar{u}_i^T a_{10}^T \bar{u}_3 + \omega_{21}^A \bar{u}_i^T a_{20}^T \bar{u}_3 + \omega_{32}^A \bar{u}_i^T a_{30}^T \bar{u}_3 = \bar{u}_i^T \{ \dot{\alpha}_1 a_1^T \bar{u}_1 + \dot{\alpha}_2 a_1^T a_2^T \bar{u}_2 + \dot{\alpha}_3 a_1^T a_2^T a_3^T \bar{u}_3 \} \quad (10)$$

($i=1, 2, 3$),

with

$$\bar{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \bar{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \bar{u}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \tilde{u}_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (11)$$

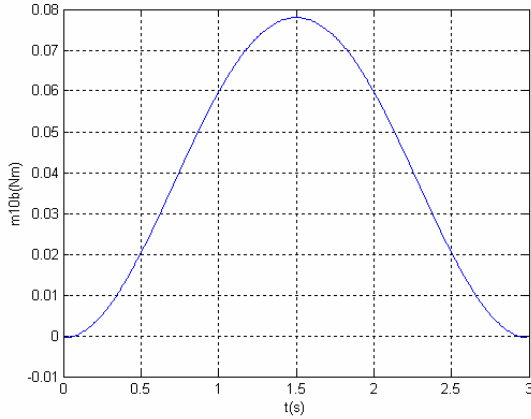


Fig. 4 Torque m_{10}^B of second actuator

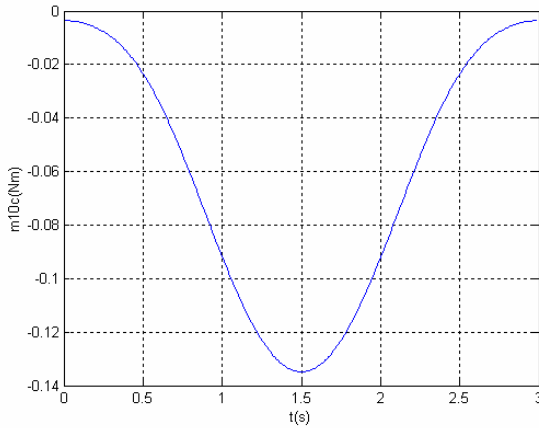


Fig. 5 Torque m_{10}^C of third actuator

These relations give the angular velocities $\omega_{10}^A, \omega_{21}^A, \omega_{32}^A$ as a function of the angular velocities $\dot{\alpha}_1, \dot{\alpha}_2, \dot{\alpha}_3$ of the end-effector. If the other two chains of the manipulator are pursued, analogous relations can then be obtained.

Expression of a *complete* Jacobian matrix of the robot is easily written in invariant form from (10). This matrix is a fundamental element for the analysis of the robot workspace and the singularity configurations.

Let us assume now that the robot has a *virtual motion* defined by the angular velocities $\omega_{10a}^{Av} = 1, \omega_{10a}^{Bv} = 0, \omega_{10a}^{Cv} = 0$. Characteristic virtual velocities expressed as function of robot's position are given by the relations of connectivity concerning the relative velocities of two independent loops $A-B$ and $B-C$:

$$\bar{u}_i^T a_{30}^T \bar{\omega}_{30a}^{Av} = \bar{u}_i^T b_{30}^T \bar{\omega}_{30a}^{Bv} = \bar{u}_i^T c_{30}^T \bar{\omega}_{30a}^{Cv} \quad (12)$$

($i=1, 2, 3$).

Some other relations of connectivity can be obtained if one considers successively that $\omega_{10b}^{Bv} = 1, \omega_{10b}^{Cv} = 0, \omega_{10b}^{Av} = 0$ and $\omega_{10c}^{Cv} = 1, \omega_{10c}^{Av} = 0, \omega_{10c}^{Bv} = 0$.

Concerning the relative angular accelerations $\varepsilon_{10}^A, \varepsilon_{21}^A, \varepsilon_{32}^A$ of the elements of first leg $A_1 A_2 A_3 G$, these are given by some other *conditions of connectivity*, obtained by deriving the relations (10); it results

$$\begin{aligned} \varepsilon_{10}^A \bar{u}_i^T a_{10}^T \bar{u}_3 + \varepsilon_{21}^A \bar{u}_i^T a_{20}^T \bar{u}_3 + \varepsilon_{32}^A \bar{u}_i^T a_{30}^T \bar{u}_3 = \\ \bar{u}_i^T \{ \ddot{\alpha}_1 a_1^T \bar{u}_1 + \ddot{\alpha}_2 a_1^T a_2^T \bar{u}_2 + \ddot{\alpha}_3 a_1^T a_2^T a_3^T \bar{u}_3 + \\ + \dot{\alpha}_1 \dot{\alpha}_2 a_1^T \tilde{u}_1 a_2^T \bar{u}_2 + \dot{\alpha}_2 \dot{\alpha}_3 a_1^T a_2^T \tilde{u}_2 a_3^T \bar{u}_3 + \\ + \dot{\alpha}_3 \dot{\alpha}_1 a_1^T \tilde{u}_1 a_2^T a_3^T \bar{u}_3 - \omega_{10}^A \omega_{21}^A a_{10}^T \tilde{u}_3 a_{21}^T \bar{u}_3 - \\ - \omega_{21}^A \omega_{32}^A a_{20}^T \tilde{u}_3 a_{32}^T \bar{u}_3 - \omega_{32}^A \omega_{10}^A a_{30}^T \tilde{u}_3 a_{10}^T \bar{u}_3 \}, \end{aligned} \quad (13)$$

where $\tilde{u}_1, \tilde{u}_2, \tilde{u}_3$ are three skew symmetric matrices associated to the unit vectors $\bar{u}_1, \bar{u}_2, \bar{u}_3$.

The following recursive relations give the angular accelerations $\bar{\varepsilon}_{k0}^A$ and the linear accelerations $\bar{\gamma}_{k0}^A$ of joints A_k [19]:

$$\begin{aligned} \bar{\varepsilon}_{k0}^A &= a_{k,k-1} \bar{\varepsilon}_{k-1,0}^A + \varepsilon_{k,k-1}^A \bar{u}_3 + \\ &+ \omega_{k,k-1}^A a_{k,k-1} \tilde{\omega}_{k-1,0}^A a_{k,k-1}^T \bar{u}_3. \\ \tilde{\omega}_{k0}^A \tilde{\omega}_{k0}^A + \tilde{\varepsilon}_{k0}^A &= a_{k,k-1} (\tilde{\omega}_{k-1,0}^A \tilde{\omega}_{k-1,0}^A + \tilde{\varepsilon}_{k-1,0}^A) a_{k,k-1}^T + \\ &+ \omega_{k,k-1}^A \omega_{k,k-1}^A \tilde{u}_3 \tilde{u}_3 + \varepsilon_{k,k-1}^A \tilde{u}_3 + \\ &+ 2 \omega_{k,k-1}^A a_{k,k-1} \tilde{\omega}_{k-1,0}^A a_{k,k-1}^T \tilde{u}_3 \\ \bar{\gamma}_{k0}^A &= a_{k,k-1} [\bar{\gamma}_{k-1,0}^A + (\tilde{\omega}_{k-1,0}^A \tilde{\omega}_{k-1,0}^A + \tilde{\varepsilon}_{k-1,0}^A) \bar{r}_{k,k-1}^A]. \end{aligned} \quad (14)$$

The relations (10) and (13) constitute the *inverse kinematics model* of the 3-RRR symmetric spherical parallel mechanisms.

IV. Dynamics simulation

Since spherical parallel robots are mostly used in applications involving high-speed motion, the dynamics of the robot has a very important effect on the required actuator torques.

Regarding the robot control, the relevant objective of a dynamic model is to determine the torques that must be exerted by the actuators in order to produce a given

trajectory of the robot. Such a model can be obtained using three main approaches, which provide the same results concerning these active moments. The first approach is the Newton-Euler classic procedure; the second one consists in the Lagrange equations and the third one is based on the principle of virtual works.

In order to establish some recursive matrix relations for the torques of the three actuators, in the inverse dynamic problem we apply the method of virtual powers.

Three electric motors A_1, B_1, C_1 , that generate three moments $\bar{m}_{10}^A = m_{10}^A \bar{u}_3$, $\bar{m}_{10}^B = m_{10}^B \bar{u}_3$, $\bar{m}_{10}^C = m_{10}^C \bar{u}_3$, which have the directions of the axes $A_1 z_1^A$, $B_1 z_1^B$, $C_1 z_1^C$, control the motion of the legs of the manipulator.

The force of inertia and the resultant moment of forces of inertia, acting on the body T_k are evaluated with the other hand, respect to the centre of the A_k joint. On the other hand, the wrench of the vectors $\bar{f}_k^*$, $\bar{m}_k^*$ designate the action of the weight $m_k \bar{g}$ and of any other external and internal applied forces at the same element of the robot. At this stage, only the torques applied on the proximal links is of interest.

The fundamental principle of the virtual powers [1], [4], [17], [20] states that a mechanism is under dynamic equilibrium if and only if the virtual power developed by all external, internal and inertia forces vanish during any general virtual displacement, which is compatible with the constraints imposed on the mechanism. Assuming that the friction at the joints is negligible, the virtual power produced by the forces of constraint at the joints is zero.

Applying the fundamental equations of the parallel robots dynamics established in compact form by Stefan Staicu [21], [24], the following matrix relation results

$$\begin{aligned} m_{10}^A = \bar{u}_3^T \{ & \bar{M}_1^A + \omega_{21a}^{Av} \bar{M}_2^A + \omega_{32a}^{Av} \bar{M}_3^A + \\ & + \omega_{21a}^{Bv} \bar{M}_{20}^B + \omega_{21a}^{Cv} \bar{M}_{20}^C \}, \end{aligned} \quad (15)$$

where one denoted

$$\begin{aligned} \bar{F}_{k0}^A &= m_k^A [\bar{\gamma}_{k0}^A + (\tilde{\omega}_{k0}^A \tilde{\omega}_{k0}^A + \tilde{\varepsilon}_{k0}^A) \bar{r}_k^{CA}] - \bar{f}_k^{*A} \\ \bar{M}_{k0}^A &= m_k^A \bar{r}_k^{CA} \bar{\gamma}_{k0}^A + \hat{J}_k^A \bar{\varepsilon}_{k0}^A + \tilde{\omega}_{k0}^A \hat{J}_k^A \tilde{\omega}_{k0}^A - \bar{m}_k^{*A} \\ \bar{F}_k^A &= \bar{F}_{k0}^A + a_{k+1,k}^T \bar{F}_{k+1}^A \quad (k=1, 2, 3) \\ \bar{M}_k^A &= \bar{M}_{k0}^A + a_{k+1,k}^T \bar{M}_{k+1}^A + \tilde{r}_{k+1,k}^A a_{k+1,k}^T \bar{F}_{k+1}^A. \end{aligned} \quad (16)$$

The relations (15) and (16) represent the *inverse dynamics model* of the spherical parallel robot. Analogous expressions we can obtain for the torques m_{10}^B and m_{10}^C .

The new dynamic approach developed here is completely general and can be used for any spherical manipulator with revolute or prismatic actuators.

The dynamics modelling of Gosselin [12] is obtained using the Newton-Euler equations of dynamic equilibrium, which are written for each of seven moving

rigid bodies of the three kinematical chains. The assumptions used here are that the joints connecting the distal links to the end-effector cannot transmit torques and consequently that they are equivalent to *spherical* joints.

Since the mechanism is over-constrained, based on the principle of virtual powers, the novel approach developed in the present paper can eliminate any assumption concerning the influence of the non-working forces applied to the distal links by the end-effector.

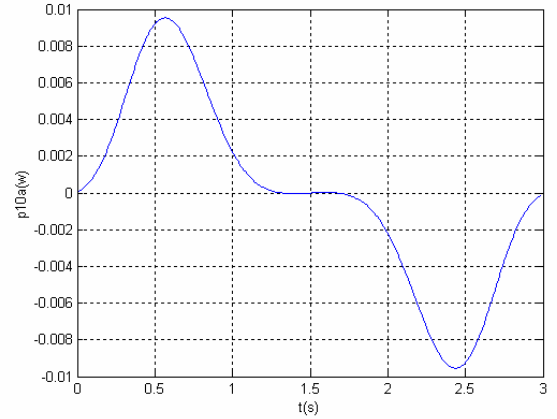


Fig. 6 Power p_{10}^A of first actuator

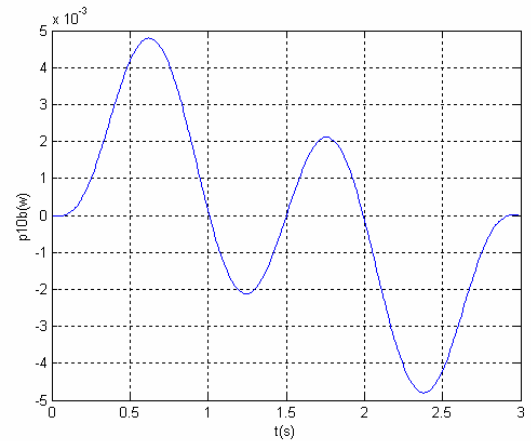


Fig. 7 Power p_{10}^B of second actuator

The Newton-Euler method, developed by Khalil in inverse dynamics of Orthoglide parallel robot [22] and Gough-Stewart platform [23], consists to apply the free-body diagram (FBD) procedure for each leg connecting the end-effector, where *all* components of joint forces and the actuated forces are unknown.

Based on the general principle of virtual powers, the present methodology can eliminate all forces of internal joints and leads immediately to a system of explicit dynamics equations for the torques or forces required by the actuators.

Consider, for example, a spherical manipulator that has the following characteristics:

$$m_1^A = m_1^B = m_1^C = m_2^A = m_2^B = m_2^C = 0.14 \text{ kg}, m_3 = 0.7 \text{ kg}$$

$$r = 0.074 \text{ m}, \Delta t = 3 \text{ s}, \alpha_1 = \frac{\pi}{36}, \alpha_2 = \frac{\pi}{18}, \alpha_3 = \frac{\pi}{12}.$$

Finally, one obtains the graphs of time-history for the torques m_{10}^A (fig.3), m_{10}^B (fig.4), m_{10}^C (fig.5) and the powers p_{10}^A (fig. 6), p_{10}^B (fig. 7), p_{10}^C (fig. 8) of the actuators.

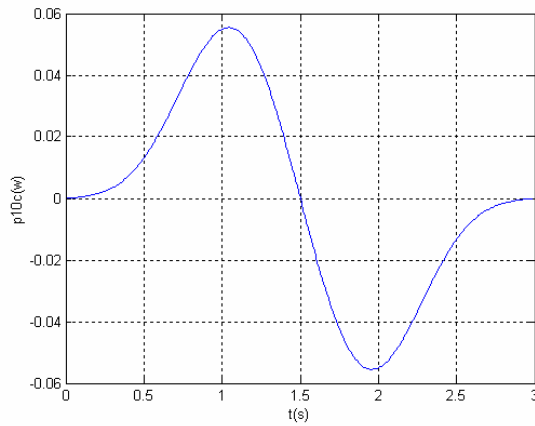


Fig. 8 Power p_{10}^C of third actuator

V. Conclusions on the advantage of the present method

Most of dynamical models based on the Lagrange formalism neglect the weight of intermediate bodies and take into consideration only the active forces or moments and the wrench of applied forces on the moving platform. The number of relations given by this approach is equal to the total number of the position variables and Lagrange multipliers inclusive. Also, the analytical calculi involved in these equations are very tedious, thus presenting an elevated risk of making errors.

The commonly known Newton-Euler method, which takes into account the free-body-diagrams of the mechanism, leads to a large number of equations with unknowns among which are also the connecting forces in the joints. Finally, the actuating torques could be obtained.

Within the inverse kinematics analysis some exact relations that give in real-time the position, velocity and acceleration of each element of the parallel robot have been established in present paper. The dynamics model takes into consideration the masses and forces of inertia introduced by the elements of the parallel mechanism.

The new approach based on the principle of virtual powers can eliminate all forces of internal joints and establishes a direct determination of the time-history evolution of torques required by the actuators. The recursive matrix relations (15) and (16) represent the explicit equations of the dynamics simulation and can easily be transformed in a model for automatic command of the spherical parallel robot.

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